

List of changes: rejected submission to current state

29 September 2026. Every item is something that was actually done, with the artefact that evidences it. Items still outstanding are listed separately at the end and are not counted as changes. Numbers come from committed result files.

A. Corrections to the evaluation protocol

#	Change	Why	Evidence
A1	Split changed from per-image to per-acquisition-field	Adjacent z-planes of one organoid were in both train and test; 50% of test images had a near-duplicate in training (median max NCC 0.950)	legacy/scientific_reports_revision , leakage audit figure; max cross-split NCC now 0.61
A2	Moved controlled experiments to public data with official splits	Private source-volume identity cannot currently be certified	results/public_study/split_manifest.json
A3	Confidence intervals recomputed with the acquisition field as resampling unit	Slices are clustered within fields; per-image bootstrap overstates precision	results/previous_private_study.csv (n_groups = 7)
A4	TNBC statistics aggregated per patient	11 patients, not 50 independent images	results/paired_statistics.csv
A5	Paired Wilcoxon with Holm correction across the comparison family	Original reported a single t-test	results/paired_statistics.csv
A6	Metrics computed at native resolution after resampling probabilities	Metrics had been computed on the 512x512 network grid	src/breastseg, regression tests
A7	Boundary metrics added (BF1@2px, surface Dice, HD95)	Paper claimed boundary awareness with no boundary metric	all result tables
A8	Equal budget for every architecture (20 epochs, same optimiser, schedule, augmentation, threshold)	Baselines had 150 epochs vs 200 for the proposed model	results/public_study/status.json, protocol hash
A9	Fixed threshold 0.5, no test-time augmentation anywhere	Manuscript claimed 7-fold TTA; code implemented 3 and the eval script called none	protocol config
A10	Test partition evaluated only after validation-based checkpoint selection	Validation had filtered low-scoring batches before selection	train_public_study.py

B. Code defects found and fixed

#	Defect	Consequence	Fix
B1	Masks thresholded at 127	508 of 855 binary 0/1 annotations read as empty, including in the sampler (8.9x under-weighting)	All nonzero labels are foreground; encoding-equivalence test
B2	Gabor kernels normalised by signed sum	Near-zero divisor for 8 of 24 kernels	L1 normalisation; finite-output tests
B3	IQR/floor filtering of validation batches plus metric smoothing	Checkpoint selection not based on the full validation set	Full validation set, unsmoothed native-grid mean Dice

#	Defect	Consequence	Fix
B4	Logits vs probabilities inferred from gradient state and maximum value	Low-magnitude positive logits thresholded incorrectly	Output contract is always logits; test at logit 0.1
B5	Unreadable images/masks silently replaced with zeros	Missing data could masquerade as true negatives	Loader fails on unreadable data or geometry mismatch
B6	16-bit images loaded through 8-bit conversion keeping the high byte	Undocumented contrast change; measured cost ~0.01 Dice	Native intensities retained to deterministic percentile normalisation
B7	Weighted-sampling flag not passed to the loader in the older export	The no-weighted-sampling ablation in that export is invalid	Explicit sampler contract and regression test
B8	Modular and monolithic loss definitions differed	Exported runs cannot be treated as sharing one objective	Single versioned Dice/BCE objective plus optional morphological boundary term
B9	Silently skipped batches; partial gradients not stepped	Effective budgets differed without being reported	Every batch used; non-finite loss or gradient stops the run

C. Claims withdrawn or corrected

#	Original claim	Current status
C1	95.1% Dice / 89.2% IoU headline	Withdrawn. Stored means exceed the recomputed means of their own 128 per-image values by exactly 0.01 for both metrics. Neither value is carried forward.
C2	Ablations show each component is essential (up to -95 Dice points)	Not reproducible. Largest measured single-component effect is under one Dice point; several components are neutral or harmful.
C3	Tversky term with recall bias; boundary-aware loss	Withdrawn. The released checkpoint used 0.7·Dice + 0.3·adaptive BCE, with neither term. New study uses an explicit Dice+BCE objective and a separately evaluated boundary term.
C4	Baselines score 0.47-0.76	Not reproduced under equal budgets. Standard baselines reach 0.931-0.938 on the corrected private protocol.
C5	"60% of images are empty"	Incorrect; artefact of mask-encoding threshold. All 855 valid images contain nuclei.
C6	EfficientNet-B7 capacity is necessary	Withdrawn. The encoder ladder is flat: B0 (6.6M params) matches B7 (68M).
C7	Clinical-grade deployment	Removed everywhere. No clinical validation was performed.
C8	"Quantum" edge enhancement; AI-generated method figure	Removed. Figures are computed by released code from real images; no quantum mechanism claimed; AI-assistance disclosure drafted.

D. New experiments run

#	Experiment	Scale	Headline outcome
D1	Corrected private-data study	38 runs, field-wise split, 3 seeds	All conditions overlap once clustered CIs are used

#	Experiment	Scale	Headline outcome
D2	Private 4-fold leave-fields-out CV	23 fields, all held out once	Pooled Dice 0.894; exposes two annotation conventions costing 0.05–0.10 Dice
D3	Matched public study (BBBC039)	21 runs, 420 epochs, 3 seeds each	Gabor bundle +0.250 Dice points vs vanilla ([0.197, 0.299], $p < 0.001$)
D4	Mechanism follow-up (SCSE-only, residual Gabor, residual Sobel, residual Gabor+boundary)	12 runs	SCSE-only and residual Sobel are worse than vanilla; residual Gabor is neutral
D5	Untuned transfer to breast histology (TNBC)	7 configurations x 11 patients	All near zero; every added component makes transfer worse
D6	Independent re-evaluation of historical checkpoints	24 checkpoints, 72 evaluations	0.46–0.71 on BBBC038; several collapse to 0.00 on BBBC039; failures retained
D7	Edge-detector controls (Sobel, Scharr, LoG, Canny, none)	private + public	No detector consistently better; network largely indifferent to the edge map
D8	Gabor parameter sensitivity (orientations, scales, sigma)	6 private runs	All within ± 0.01 Dice
D9	Encoder ladder (ResNet-34/50, EfficientNet-B0/B3/B5/B7)	6 private runs	Flat across a tenfold parameter range

E. Infrastructure and reporting

#	Change
E1	New corrected implementation under src/breastseg; historical exports preserved unedited as evidence under legacy/
E2	Protocol hashing: a completed run is skipped only if its protocol hash matches; partial runs stop for inspection
E3	Scientific regression tests (tests/): mask encodings, empty-mask Dice conventions, logit handling, sampler behaviour, kernel finiteness
E4	All reports and the manuscript draft generated from committed CSV/JSON, so abstract, tables and text cannot diverge
E5	6,060 evaluated masks retained locally; per-image scores and checkpoint hashes committed
E6	Qualitative figures produced from real images, predictions and logs; cases selected by rank, not appearance
E7	Dataset licensing and attribution documented (reports/datasets_and_sources.md)
E8	Reviewer-response tracker mapping all 20 Reviewer 1 items and Reviewer 2's concerns to evidence or remaining work

F. Still outstanding — not changes, blockers

#	Item	Why it matters
F1	Private source-volume/acquisition identifiers	Field grouping is reconstructed from frame geometry, not metadata; biological independence is unproven
F2	Annotation protocol, annotator count, inter-rater agreement	Reviewer 1 item 10; cannot be reconstructed retrospectively. CV shows two conventions exist, which makes this more urgent, not less

#	Item	Why it matters
F3	Public code and weight release, inherited-code licensing, data-release permission	Reviewer 1 item 3; the repository is currently private, which does not satisfy it
F4	Confirmatory protocol on genuinely unexamined data	Current public results are exploratory: the test labels have now been inspected
F5	Matched-data modern baselines, including a fine-tuned foundation model	Cellpose-SAM zero-shot is context, not an equal-data comparison
F6	Author list, contributions, funding, ethics and AI-use disclosure	Required before any submission
F7	A defensible contribution	If the edge branch does not improve matched baselines, a superiority paper is not justified